Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		copper(II) chloride (1) accept copper chloride / $CuCl_2$ cathode (1) electrolysis (1)	3			3		3
		(ii)		$CuCl_2$		1		1		
		(iii)		$Cu - 2e^- \longrightarrow Cu^{2+}$ ☐ $Cu + 2e^- \longrightarrow Cu^{2+}$ ☐ $Cu^{2+} + 2e^- \longrightarrow Cu$ ☑ $Cu^{2+} - 2e^- \longrightarrow Cu$ ☐		1		1		
	(b)	(i)		to release the oxygen gas from the aluminium oxide ☐ to allow the aluminium ions and oxide ions to move ☑ to allow the aluminium to leave the cell ☐ to speed up the process ☐	1			1		

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(ii)		$2\,Al_2O_3 \longrightarrow \boxed{4}\,Al + 3\,O_2$		1		1	1	
		(iii)		landfill sites ☐ power stations ☑ limestone quarries ☐ oil refineries ☐ coastal ports ☑ coal mines ☐ award (1) for each correct answer deduct (1) for each additional tick if more than two ticks	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			award (1) each for any **two** of following • low density • does not corrode • ductile neutral answers malleable / durable / does not tarnish / weather resistant do not accept strong / high melting point / unreactive	2			2		
				Question 2 total	**8**	**3**	**0**	**11**	**1**	**3**